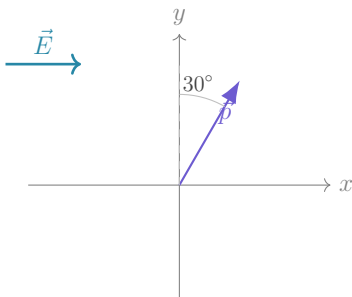


Electrostatics Review — Flex Day, Sep 30

Part A — Dipole: Torque, PE, and Motion

1. A uniform field $\vec{E} = 40 \text{ N/C}$ points in $+\hat{x}$. A dipole is made of two $\pm 3.0 \text{ nC}$ charges separated by $d = 2.0 \text{ mm}$, with its moment $\vec{p}$ tilted 30° to the right of the y -axis (i.e. 60° from $\vec{E}$).



(a) What is the torque on the dipole, as a vector (magnitude and direction)?

(b) What is the dipole's potential energy in this orientation?

(c) The dipole is released from rest and rotates until $\vec{p}$ aligns with $\vec{E}$ (i.e. $\theta \rightarrow 0$). Taking each charge's mass to be $m = 1.2 \times 10^{-14} \text{ kg}$, what speed does each charge have at the instant $\theta = 0$? *Hint: both charges move on a circle of the same radius about the center, so they share the same speed — the dipole's lost PE becomes the total KE of both charges together.*

Part B — Even Steps in Potential, Uniform Field

2. A parallel-plate capacitor has plate separation 6.0 mm and a uniform field $E = 3000 \text{ N/C}$ between the plates, pointing from the $+$ plate to the $-$ plate.

(a) Draw the field lines between the plates (evenly spaced, pointing from $+$ to $-$).

(b) Draw four equally-spaced equipotential lines between the plates, and label the potential of each one — take $V = 0$ at the negative plate.

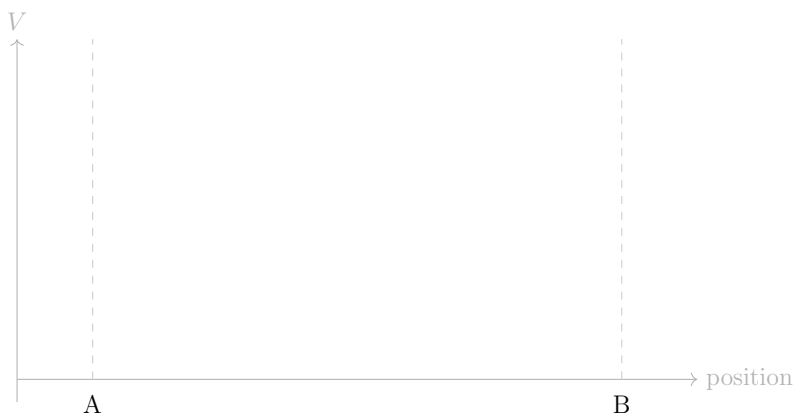


(plates are 6.0 mm apart — not drawn to scale)

(c) A $+4.0 \text{ nC}$ charge moves from the positive plate to a point 1.5 mm away (one quarter of the way across). How much potential energy does it lose?

Part C — V Doesn't Have to Change Evenly to Know ΔU

3. A charge moves from point A, where $V_A = +10$ V, to point B, where $V_B = +40$ V — but the potential along the way is *not* uniform: it rises and dips in between (like a bumpy hill, not a smooth ramp). Sketch a plausible bumpy $V(x)$ between A and B below, consistent with those two endpoint values.

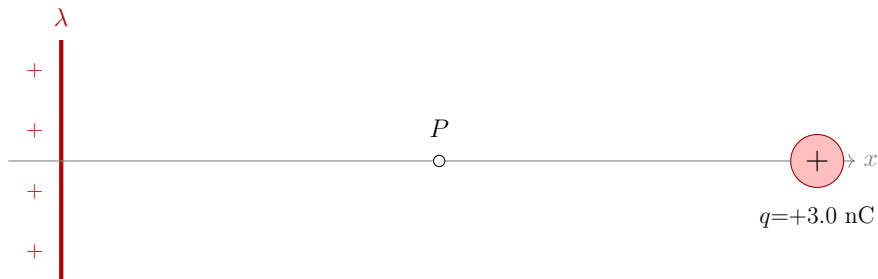


(a) A $+2.0$ nC charge moves from A to B along this bumpy path. What is ΔU ?

(b) Does the bumpiness of the path matter to your answer in (a)? Why or why not? (Think about the ball-on-a-hill analogy — does it matter if the hill between the start and end points is smooth or bumpy?)

Part D — Superposition with Continuous Distributions

4. An infinite line of charge with density $\lambda = +2.0 \times 10^{-8}$ C/m runs along the y -axis. A point charge $q = +3.0$ nC sits on the x -axis at $x = 10$ cm. Point P sits on the x -axis at $x = 5.0$ cm — between the line and the point charge.



(a) Find the field at P due to the line alone.

(b) Find the field at P due to the point charge alone.

(c) Add them as vectors (careful with directions!) to find the net field at P .

5. An infinite plane of charge with density $\eta = +1.0 \times 10^{-9} \text{ C/m}^2$ lies in the yz -plane. A point charge $q = +5.0 \text{ nC}$ sits on the x -axis at $x = 2.0 \text{ cm}$, on the same side as point P at $x = 1.0 \text{ cm}$.
(a) Find the field at P due to the plane alone, and due to the point charge alone.

(b) Add them as vectors to find the net field at P . Which source dominates, and does that surprise you given how differently each one's field falls off with distance?